

ZUWARA, Libyan Arab JamahiriyaWMO: **620070**Lat: **32.883N**Lon: **12.083E**Elev: **4**StdP: **101.28**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	8.9	-0.5	3.6	13.8	1.0	4.1	14.0	12.5	15.7	11.1	15.6	3.9	240	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	35.8	23.1	33.2	24.3	31.5	25.1	28.2	31.2	27.4	30.3	26.8	29.8	4.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.5	23.4	30.5	26.6	22.2	29.7	26.0	21.3	29.2	90.8	31.3	86.9	30.3	84.1	29.9	33.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.4	9.8	8.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	14.8	15.0	16.9	18.9	22.1	24.9	27.4	28.5	27.1	23.8	18.9	15.4
	DBStd	5.55	1.87	2.58	3.18	2.96	2.98	2.47	1.76	1.64	1.99	2.81	2.97	2.97
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	431	111	100	68	21	2	0	0	0	0	1	28	102
	CDD10.0	4083	149	140	213	267	375	448	538	575	514	428	268	168
	CDD18.3	1472	2	7	22	39	118	198	280	317	264	170	45	11
	CDH23.3	13260	13	52	176	253	533	1293	2937	3798	2757	1188	194	64
CDH26.7	4309	1	13	82	126	228	311	893	1469	841	294	37	14	
Wind	WSAvg	4.4	3.7	4.2	4.4	5.0	4.9	4.8	4.3	4.5	4.8	4.1	3.8	4.0
Precipitation	PrecAvg	240	35	18	18	13	6	1	0	1	21	42	38	46
	PrecMax	454	162	63	80	83	73	5	1	10	129	165	190	172
	PrecMin	40	2	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	98	31	18	21	18	13	1	0	2	30	44	46	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	28.6	33.5	35.9	39.5	37.6	37.0	38.4	38.8	35.5	29.8	28.4
		MCWB	14.6	17.9	20.9	20.1	25.0	22.2	24.9	24.2	22.6	22.1	19.1	17.1
	2%	DB	22.1	24.4	27.8	29.7	32.5	32.0	32.6	33.7	33.2	31.7	27.4	24.9
		MCWB	14.2	15.5	17.4	19.1	21.1	23.5	25.4	26.3	24.9	22.9	18.0	16.6
	5%	DB	20.0	21.4	23.4	25.2	27.6	29.5	31.0	32.0	31.1	29.2	25.4	21.9
		MCWB	13.1	14.6	16.1	17.8	20.5	23.4	25.6	26.7	25.5	23.8	19.3	15.6
	10%	DB	18.6	19.2	20.9	22.6	25.2	27.8	30.0	31.0	30.0	27.8	23.8	20.1
		MCWB	13.7	13.9	16.1	17.5	20.4	23.1	25.2	26.3	25.1	23.3	18.9	15.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.9	20.0	21.5	23.2	27.8	26.9	28.6	30.4	28.4	27.6	23.2	20.2
		MCDB	20.3	27.3	27.7	29.3	36.1	31.7	32.1	33.9	31.2	30.2	25.7	23.5
	2%	WB	16.2	17.2	18.9	20.4	23.1	25.6	27.1	28.6	27.5	26.3	21.8	18.2
		MCDB	19.2	21.4	24.0	26.4	28.7	29.3	30.2	31.3	30.3	28.6	24.4	21.7
	5%	WB	15.3	15.8	17.4	19.0	21.7	24.6	26.5	27.7	26.7	25.2	20.7	17.1
		MCDB	18.2	19.3	21.1	23.6	25.3	27.8	29.6	30.4	29.7	27.7	23.8	20.6
	10%	WB	14.6	14.9	16.7	18.2	20.9	23.9	26.0	27.1	26.1	24.1	19.5	15.9
		MCDB	17.5	18.0	19.8	21.6	24.3	26.8	29.1	30.0	29.1	27.1	23.1	19.4
Mean Daily Temperature Range	5% DB	MDBR	7.7	7.9	7.6	6.6	6.0	5.3	5.1	5.2	5.6	7.0	8.3	8.3
		MCDBR	11.2	12.4	13.5	12.9	11.7	9.0	7.4	7.3	8.1	9.6	10.3	11.2
		MCWBR	6.9	7.6	6.8	6.3	6.2	5.2	5.0	5.2	5.4	6.5	6.5	7.3
	5% WB	MCDBR	8.0	9.9	9.5	9.6	7.8	7.0	5.9	6.2	6.0	7.4	8.1	9.1
		MCWBR	6.4	7.4	6.4	5.8	5.4	5.0	4.9	4.9	4.9	6.0	6.7	7.1
Clear-Sky Solar Irradiance	taub	0.389	0.414	0.495	0.551	0.553	0.501	0.507	0.503	0.528	0.484	0.435	0.398	
	taud	2.192	2.108	1.893	1.789	1.815	1.964	1.967	1.990	1.931	2.003	2.079	2.158	
	Ebn at Noon	804	824	781	757	759	798	790	787	743	740	742	765	
	Edh at Noon	117	141	188	218	214	184	183	176	179	155	130	115	
All-Sky Solar Radiation	RadAvg	3.12	3.99	5.16	6.23	6.93	7.55	7.64	6.88	5.59	4.30	3.30	2.74	
	RadStd	0.16	0.20	0.25	0.32	0.32	0.27	0.17	0.14	0.21	0.24	0.17	0.20	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)		+0.30	N/A	N/A	N/A	N/A	N/A	N/A	-53	+116